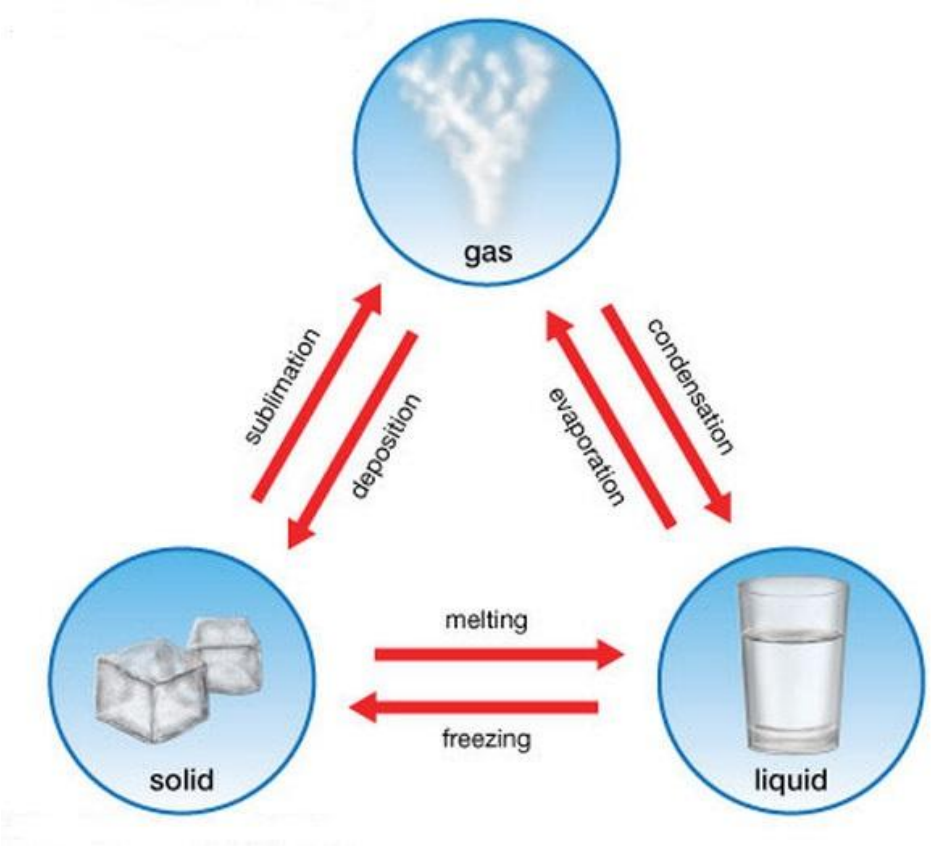
Physical & Chemical Changes of Matter

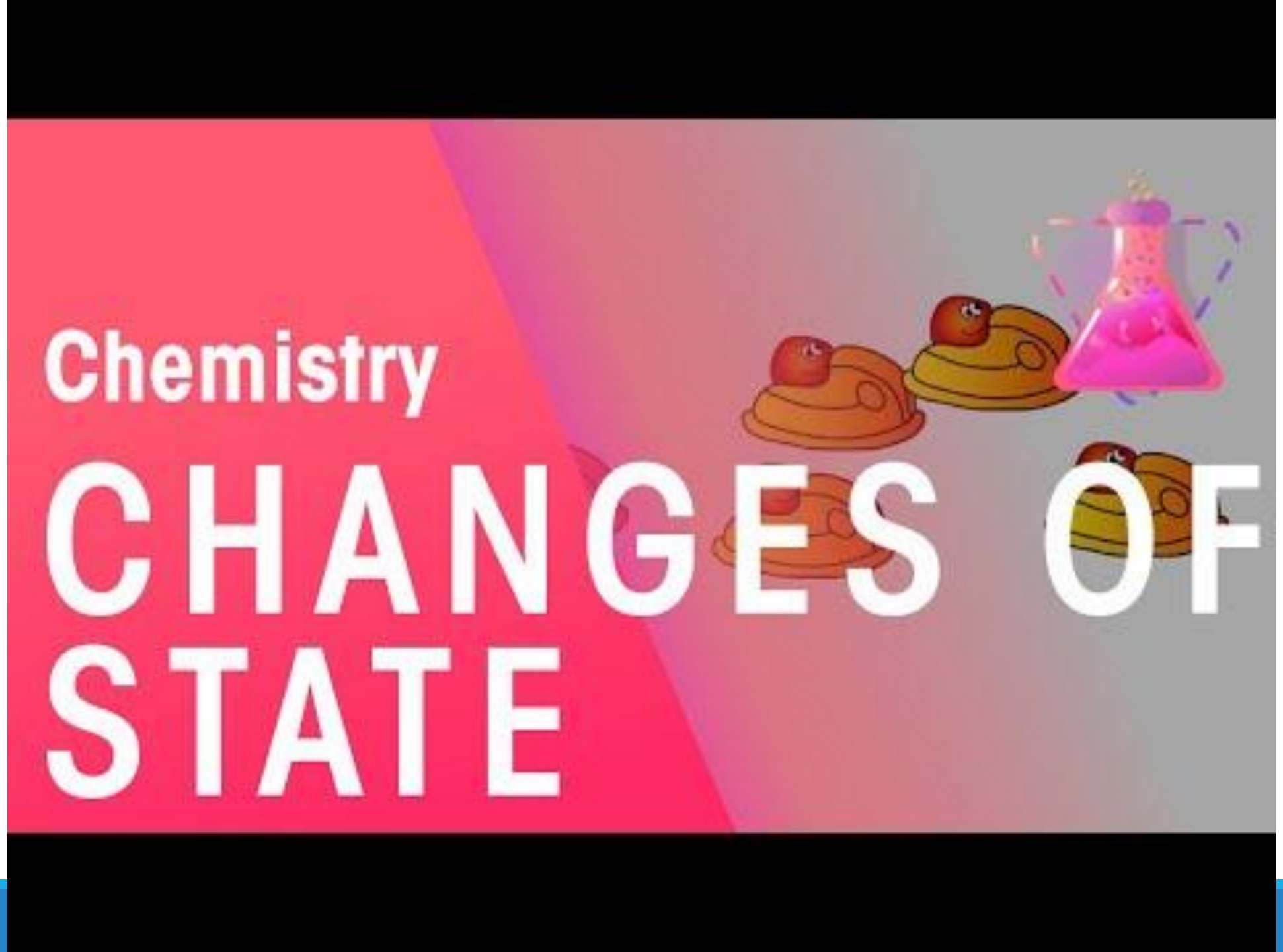
Physical Changes

- If you tear a piece of **paper** in half, it is still **paper**, only in smaller pieces
- A **physical** change has taken place – the substance remains the **same**, only changes **form**

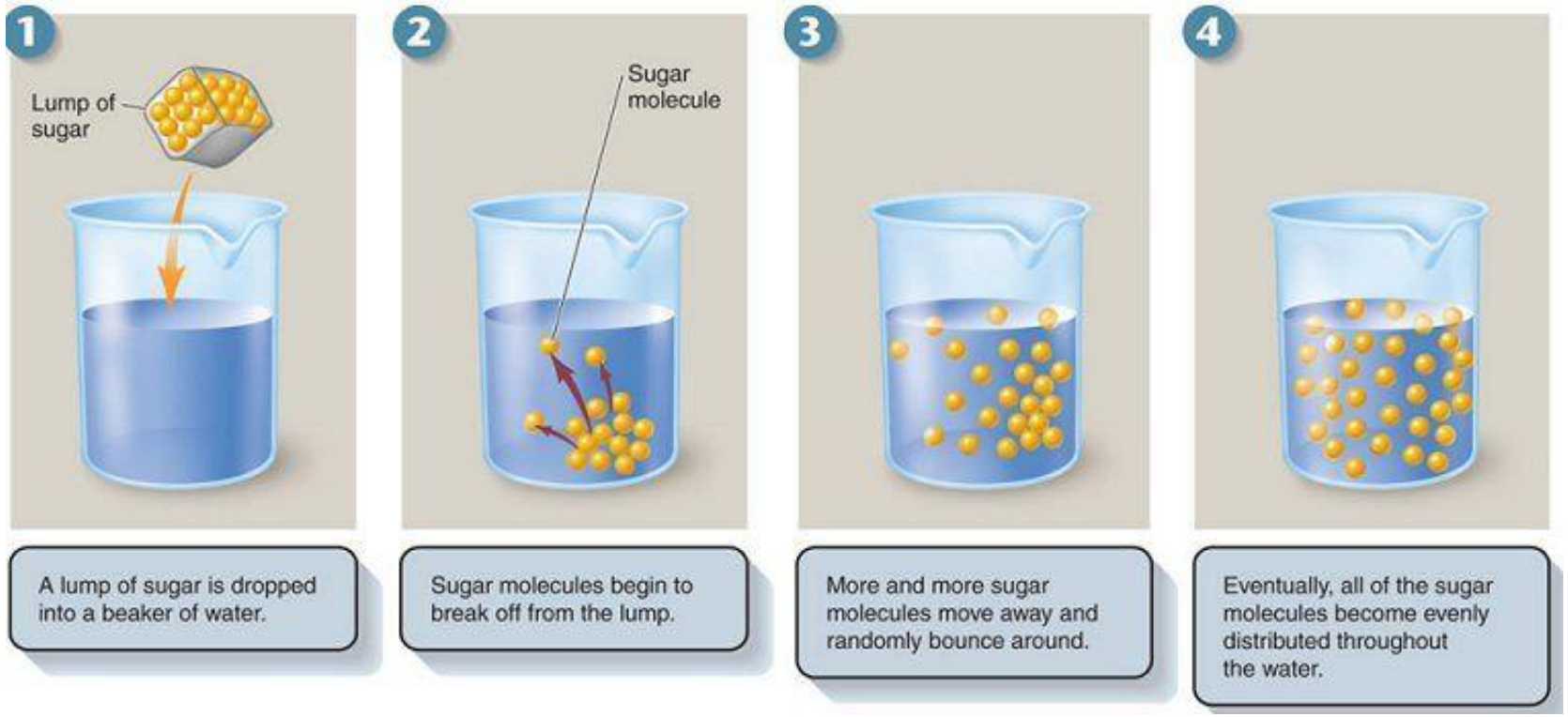
Changes of State

- Changes of state are also **physical** changes
- No matter what **state** it is in, it is still the **same** substance




Chemistry
CHANGES OF
STATE

Dissolving



- Dissolving is also a **physical** change
- If you **evaporate** the water, the sugar would be left behind

Chemical Changes

- In a chemical change, a **new** substance is formed
- **Burning** a candle is an example of a chemical change – **carbon dioxide** , **water** vapour, **heat** and **light** are being produced

Chemical Change Clues

The following clues will help you identify if a chemical change has occurred.

You need to have <u>at least two of these clues</u> present to confirm that the change is a chemical one.

Chemical Change Clues

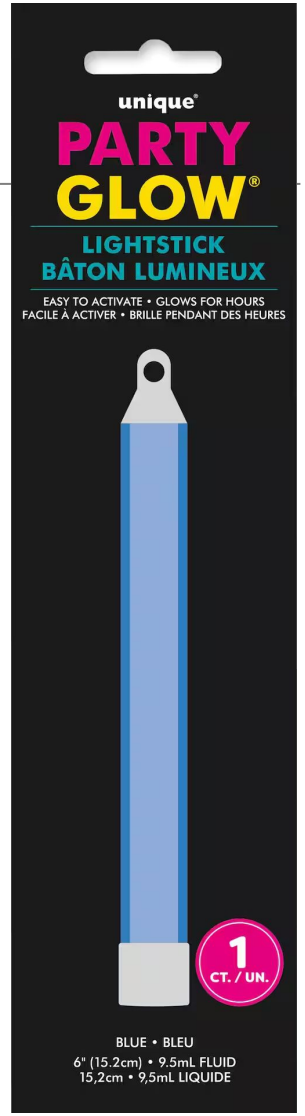
- A new **colour** appears

Chemical Change Clues

- **Bubbles** of **gas** form

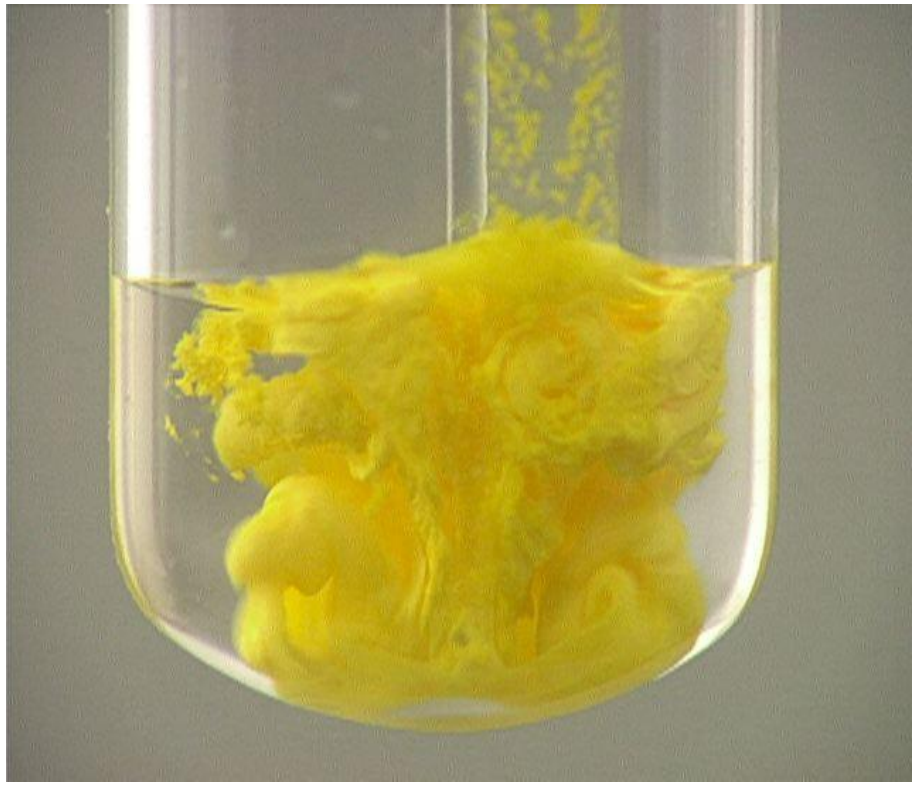
Chemical Change Clues

- **Light** and / or **heat** are given off.



Chemical Change Clues

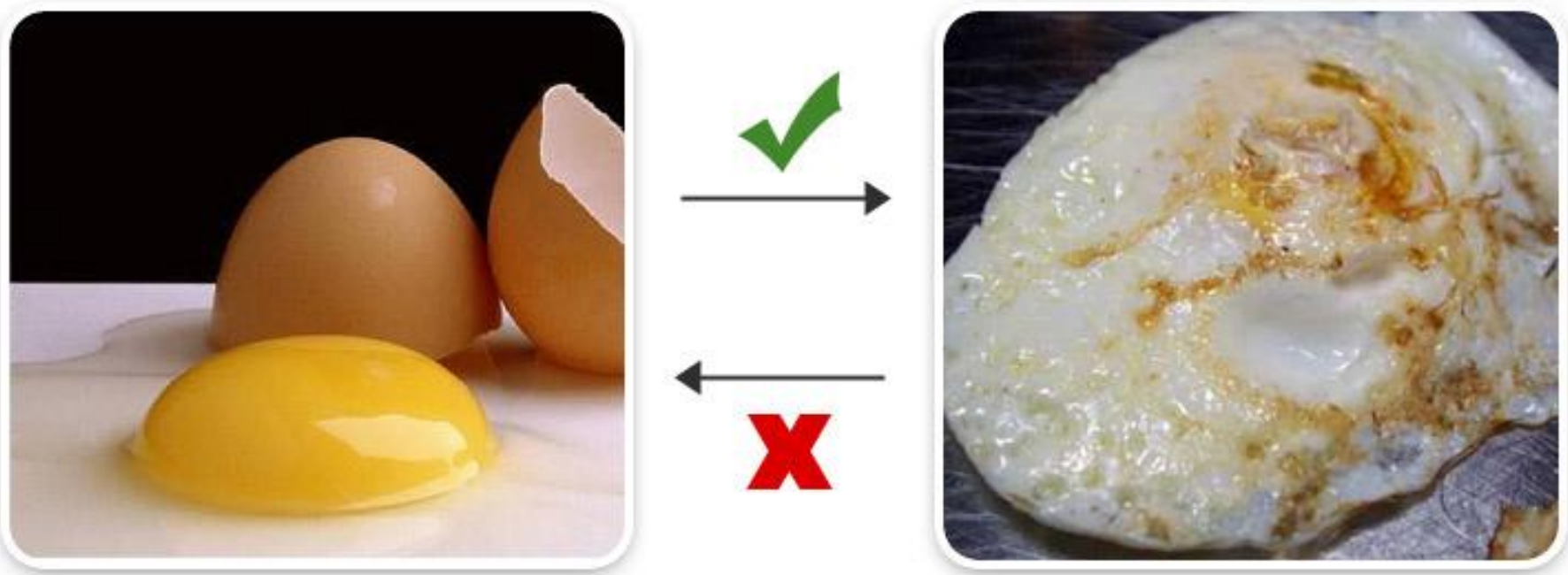
- A **precipitate** (solid) forms when two **solutions** are mixed together



$Pb(NO_3)_2$(aq) and KI (aq)

Chemical Change Clues

- The change is **difficult** , if not **impossible** to reverse



EPISODE 19.2
KITCHEN CHEMICAL CHANGES
KIDS

Your turn

- Activity: Chemical or Physical Change? (next class)
- Physical or Chemical Change? (Practice)